

Fair robustness starts with balanced mistakes between classes.

1 Balance the directions of error

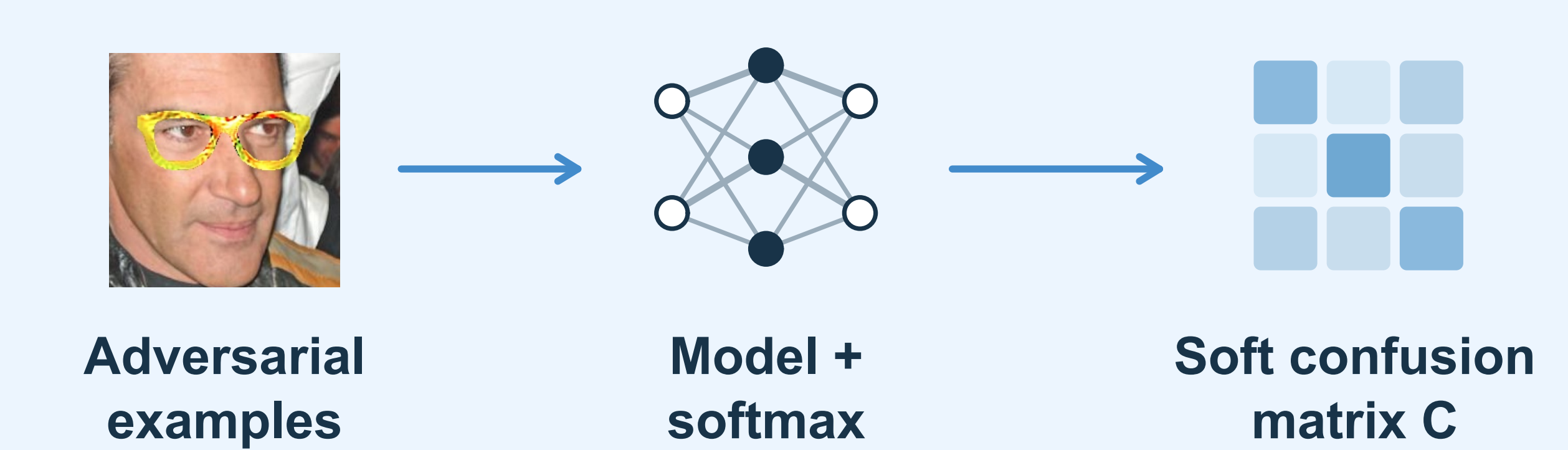
Similar classes can be harder to distinguish.
Seek symmetric confusion, rather than perfect class parity.



Both examples: 60% average robust accuracy

2 Add a symmetry objective

Use soft predictions to compare every class pair.
The full path remains differentiable.



$$\text{Pair penalty} = \frac{|a - b|}{a + b + \epsilon} \times (a + b) \quad \begin{matrix} a = C[i,j] \\ b = C[j,i] \end{matrix}$$

Sum over $i < j$

Weighted loss: clean + adversarial + symmetry

3 Protect the weakest classes

PubFig with siblings · untargeted eyeglass attack
Higher accuracy and a smaller class gap.



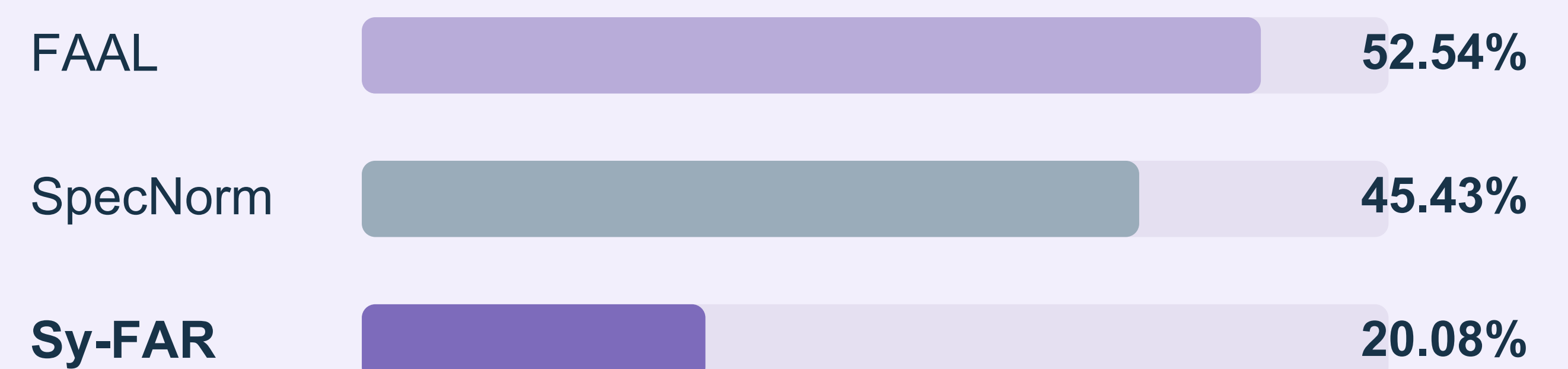
3.5× worst-class accuracy

40.9% smaller class gap

4 Reduce target concentration

Some identities attract a disproportionate share of errors.
Sy-FAR reduces this impersonation risk.

Largest share of errors into one target ↓

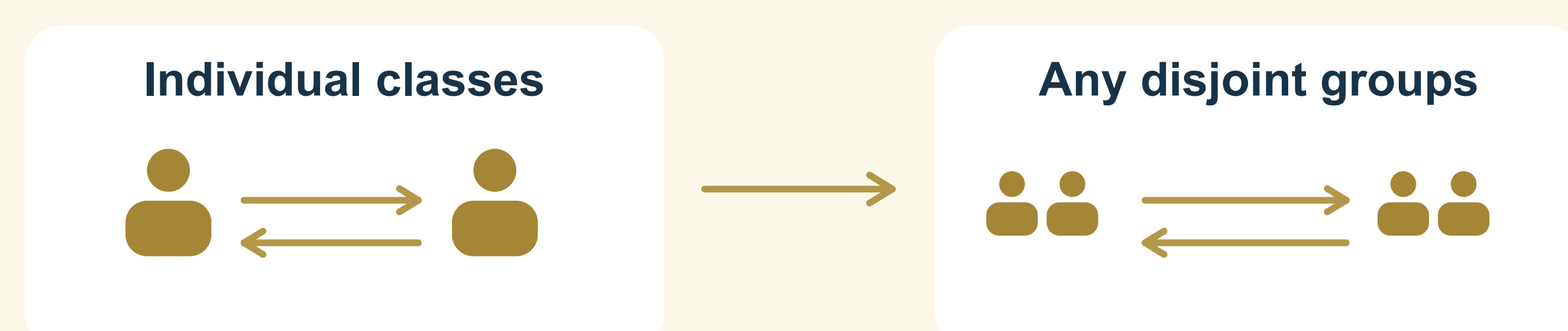


PubFig with siblings · share of all misclassifications

55.8% lower than SpecNorm

5 Extend fairness to groups

Class symmetry implies group symmetry for every partition.
No group labels are used during training.



Theorem uses pair-averaged intergroup confusion.

Gender robustness gap on PubFig with siblings ↓

36 pp
FAAL

32 pp
SpecNorm

4 pp
Sy-FAR

6 Keep training practical

Small overhead on adversarial training.
More consistent results across ten independent runs.

Nearly 5×
faster than FAAL

>2×
faster than SpecNorm

PubFig · VGG-16 · one NVIDIA TITAN Xp

Lowest run-to-run variability in:

- ✓ Robust accuracy
- ✓ Worst-class accuracy
- ✓ Class accuracy gap

$O(K^2)$ pairwise work

Vectorizable on GPU

EVALUATION

2 vision tasks · 3 architectures · 10 runs

PubFig / siblings + CIFAR-10 / 100
VGG-16 · ViT · PreAct-ResNet18
Eyeglass attacks + AutoAttack · 80/10/10 split

SCOPE & NEXT STEPS

Balanced, finite-label vision settings.

Next: imbalanced or noisy data, open-world classes, language and generative models.

pp = percentage points



Full paper